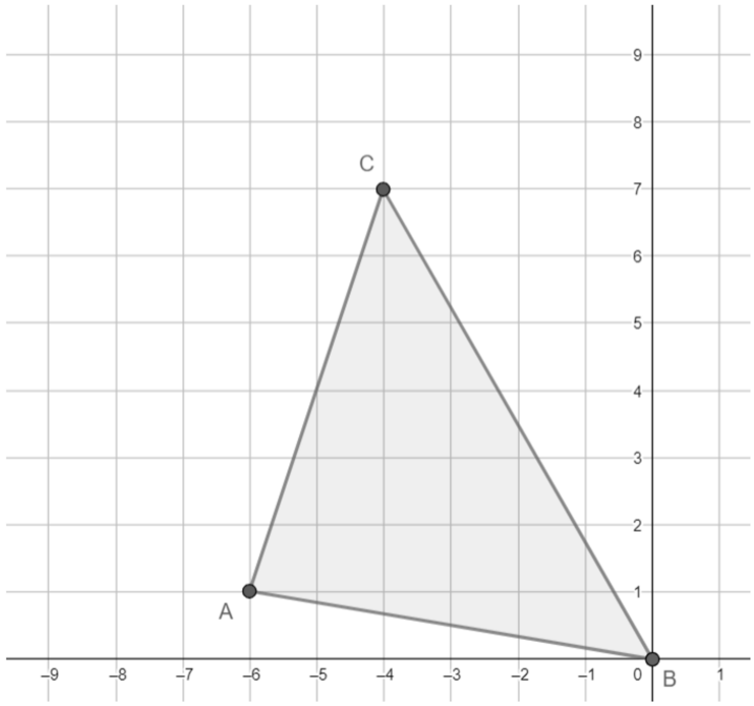
Geometry Name____________________
Unit 13 Date_____________________
Lesson 7 Homework Period___________________

Use the following information for questions 1-4. Triangle ABC has vertices at $A(-6,1)$, $B(0,0)$, and $C(-4,7)$.



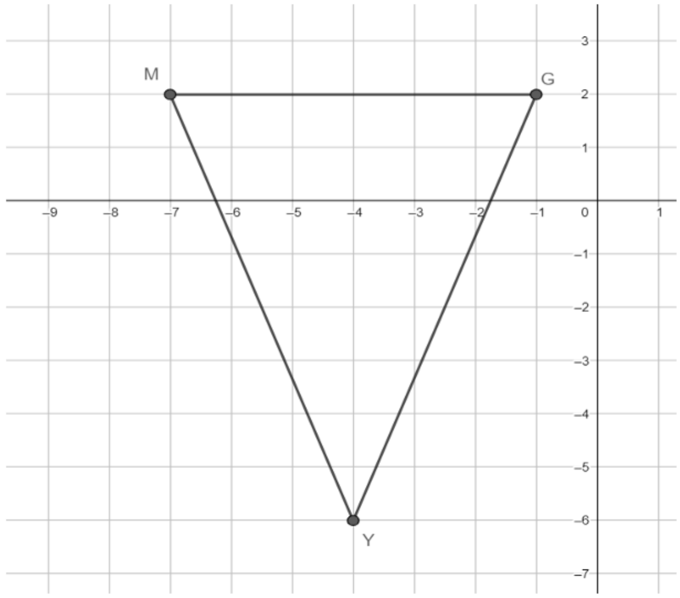
1. What is the average of the x-coordinates? Round to the nearest thousandth if necessary.

2. What is the average of the y-coordinates? Round to the nearest thousandth if necessary.

3. What are the coordinates of the centroid?

4. Verify the coordinates of the centroid by drawing the median of each side.

Use the following information for question 5. Triangle GMY has vertices at $G(-1,2)$, $M(-7,2)$, and $Y(-4,-6)$.



5. What are the coordinates of the centroid of triangle GMY?

6. $\triangle KLM$ has vertices at $K(7,2)$, $L(6,9)$, and a centroid at $(5,6)$. What are the coordinates of vertex M?

Keziah is finding the centroid of $\triangle ALG$ with vertices at $A(-1,3)$, $L(2,0)$, and $G(-4,-6)$.

Use the following information for questions 7-8. Her work is shown.

Keziah's Work	
$x = \frac{1+2+4}{3}$ $x = \frac{7}{3}$	$y = \frac{3+0+6}{3}$ $y = \frac{9}{3} = 3$
Centroid $\left(\frac{7}{3}, 3\right)$	

7. What is Keziah's mistake?

8. What are the coordinates of the centroid of $\triangle ALG$?